and birds nesting. According to their data, many insects whose life cycles are governed by air temperature (thermophilic insects, such as butterflies and bark beetles) now have a longer breeding season, enabling them to produce extra generations each year. For example, some butterflies that previously had two generations now have three.

In Spain, botanists studied data for 29 species of plants. They found that in 2003, leaves first appeared 4.8 days earlier on average than in 1943; flowers first bloomed 5.9 days earlier; trees produced fruit 3.2 days earlier; and leaves died 1.2 days later. In the UK, the evidence was even more dramatic: across 53 plant species, leaves, flowers, and fruits appeared almost six days earlier in 2005 than they had done in 1976. Similarly, the fruiting season of 315 different kinds of fungi studied in Britain lengthened from 33 to 75 days in the second half of the 20th century.

Longer plant growing seasons sound like good news, but warmer temperatures create problems as well as advantages. Not all insects are welcome, and shorter, milder winters kill fewer dormant insects, some of which may consequently undergo explosive population increases and produce damaging infestations. Warmer springs allow pine sawflies, whose larvae eat pine needles, to develop too rapidly for the birds and parasites that feed on them to keep their numbers in check. Out of control, the sawflies strip trees of their needles and stunt their growth.